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$\boxed{\text{Quantum Algorithms HW1}}$

① a) $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ Hermitian $\sigma_2 = \sigma_2^\dagger = (\sigma_2^*)^T$

$$\sigma_2^* = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \Rightarrow \boxed{\sigma_2^\dagger = (\sigma_2^*)^T = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_2} \quad \square$$

b) Anti-Sym. $A^T = -A$

$$\sigma_2^T = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \quad -\sigma_2 = -\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

so $\boxed{\sigma_2^T = -\sigma_2 \text{ so Antisymmetric}}$

c) $\sigma_x x = \lambda x$ $\det(\sigma_2 - \lambda I) = 0$

$$\det\begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = (-\lambda)(-\lambda) - (-i)(i) = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$$

for $\lambda = 1$: $(\sigma_2 - 1 \cdot I)x_1 = 0 \Rightarrow \begin{pmatrix} -1 & -i \\ i & -1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

where $u_1 = -iu_2$

$u_1 = 1,\ u_2 = i$

so normalizing vector $\begin{pmatrix} 1 \\ i \end{pmatrix}$: $\boxed{x_1 = \frac{1}{\sqrt{1^2 + |i|^2}}\begin{pmatrix} 1 \\ i \end{pmatrix} = 1/\sqrt{2}\begin{pmatrix} 1 \\ i \end{pmatrix}}$

for $\lambda = -1$: $(\sigma_2 + 1 \cdot I)x_2 = 0 \Rightarrow \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

where $v_1 = iv_2$

$v_2 = -i,\ v_1 = 1$

so normalizing vector $\begin{pmatrix} 1 \\ -i \end{pmatrix}$: $\boxed{x_2 = \frac{1}{\sqrt{1^2 + (-i)^2}}\begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}}$

d) $P = (x_1\ x_2) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$

P is unitary so conjugate transpose is $P^{-1} = P^{\dagger} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$

verify transform. $P^{-1}\sigma_2 P$

$$\sigma_2 P = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ i & i \end{pmatrix}$$

$$P^{-1}\sigma_2 P = \frac{1}{2}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}\begin{pmatrix} 1 & -1 \\ i & i \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

$$\boxed{P^{-1}\sigma_2 P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} \quad \square$$

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2 a) $Hx_i = \lambda_i x_i$

$$x_i^{\dagger} H x_i = x_i^{\dagger}(\lambda_i x_i) = x_i^{\dagger}\lambda_i x_i$$

take conjugate transpose both sides

$$(x_i^{\dagger} H x_i)^{\dagger} = (x_i^{\dagger}\lambda_i x_i)^{\dagger}$$

use property $(AB)^{\dagger} = B^{\dagger}A^{\dagger}$ and $H^{\dagger} = H$

$$x_i^{\dagger} H^{\dagger}(x_i^{\dagger})^{\dagger} = \lambda_i^{*} x_i^{\dagger}(x_i^{\dagger})^{\dagger}$$

$$x_i^{\dagger} H x_i = \lambda_i^{*} x_i^{\dagger} x_i$$

Equating the expressions for $x_i^{\dagger} H x_i$:

$$\lambda_i x_i^{\dagger} x_i = \lambda_i^{*} x_i^{\dagger} x_i$$

$$(\lambda_i - \lambda_i^{*}) x_i^{\dagger} x_i = 0$$

x_i is non-zero eigenvector $x_i^{\dagger} x_i = \|x_i\|^2 > 0$

divide by $x_i^{\dagger} x_i$

$$\lambda_i - \lambda_i^{*} = 0 \Rightarrow \underline{\lambda_i = \lambda_i^{*}}$$

since it equals complex conjugate, eigenvalue λ_i is real $\square$

b) two eigenvectors x_i and x_j w/ corresponding $\lambda_i \neq \lambda_j$:

$$Hx_i = \lambda_i x_i \qquad Hx_j = \lambda_j x_j$$

$\downarrow$ multiply by $x_j^\dagger$

$$x_j^\dagger H x_i = x_j^\dagger(\lambda_i x_i) = \lambda_i x_j^\dagger x_i$$

take conjugate transpose for equation for x_j:

$$(Hx_j)^\dagger = (\lambda_j x_j)^\dagger \Rightarrow x_j^\dagger H^\dagger = \lambda_j^* x_j^\dagger$$

Using facts $H^\dagger = H$ and $\lambda_j^* = \lambda_j$ (λ_j is real)

$$x_j^\dagger H = \lambda_j x_j^\dagger$$

multiply by x_i

$$x_j^\dagger H x_i = \lambda_j x_j^\dagger x_i$$

subtract two expression ~~as~~ obtained for $x_j^\dagger H x_i$:

$$\lambda_i x_j^\dagger x_i - \lambda_j x_j^\dagger x_i = 0$$

$$(\lambda_i - \lambda_j) x_j^\dagger x_i = 0$$

Since eigenvalues are non-degenerate ($\lambda_i \neq \lambda_j$), the factor $(\lambda_i - \lambda_j) \neq 0$

Thus,

$$\boxed{x_j^\dagger x_i = 0 \quad \text{for } i \neq j}$$